# torch.round#

https://pytorch.org/docs/stable/generated/torch.round.html#torch.round

**Description:**

Rounds elements of input to the nearest integer. For integer inputs, follows the array-api convention of returning a copy of the input tensor. The return type of output is same as that of input's dtype. This function implements the “round half to even” to break ties when a number is equidistant from two integers (e.g. round(2.5) is 2). When the :attr:`decimals` argument is specified the algorithm used is similar to NumPy's around. This algorithm is fast but inexact and it can easily overflow for low precision dtypes. Eg. round(tensor([10000], dtype=torch.float16), decimals=3) is inf.

## Function Signature

```
torch.round(input, *, decimals=0, out=None) → Tensor#
Parameters
Keyword Arguments
```

## Parameters

### Keyword

out (Tensor, optional) – the output tensor.

## Code Examples

```
>>> torch.round(torch.tensor((4.7, -2.3, 9.1, -7.7)))
tensor([ 5.,  -2.,  9., -8.])

>>> # Values equidistant from two integers are rounded towards
the
>>> #   the nearest even value (zero is treated as even)
>>> torch.round(torch.tensor([-0.5, 0.5, 1.5, 2.5]))
tensor([-0., 0., 2., 2.])

>>> # A positive decimals argument rounds to the to that decimal
place
>>> torch.round(torch.tensor([0.1234567]), decimals=3)
tensor([0.1230])

>>> # A negative decimals argument rounds to the left of the
decimal
>>> torch.round(torch.tensor([1200.1234567]), decimals=-3)
tensor([1000.])
```

## Notes & Warnings

NOTE

Note This function implements the “round half to even” to break ties when a number is equidistant from two integers (e.g. round(2.5) is 2). When the :attr:`decimals` argument is specified the algorithm used is similar to NumPy’s around. This algorithm is fast but inexact and it can easily overflow for low precision dtypes. Eg. round(tensor([10000], dtype=torch.float16), decimals=3) is inf.

NOTE

See also torch.ceil(), which rounds up. torch.floor(), which rounds down. torch.trunc(), which rounds towards zero.

# Detailed Documentation

## Documentation

Rounds elements of input to the nearest integer.

For integer inputs, follows the array-api convention of returning a copy of the input tensor. The return type of output is same as that of input’s dtype.

This function implements the “round half to even” to break ties when a number is equidistant from two integers (e.g. round(2.5) is 2).

When the :attr:`decimals` argument is specified the algorithm used is similar to NumPy’s around. This algorithm is fast but inexact and it can easily overflow for low precision dtypes. Eg. round(tensor([10000], dtype=torch.float16), decimals=3) is inf.

torch.ceil(), which rounds up. torch.floor(), which rounds down. torch.trunc(), which rounds towards zero.

input (Tensor) – the input tensor.

decimals (int) – Number of decimal places to round to (default: 0). If decimals is negative, it specifies the number of positions to the left of the decimal point.

out (Tensor, optional) – the output tensor.